

Markscheme

Mathematics: analysis and approaches
Practice question bank

69. (a)

The 2nd term of a geometric sequence is 6 and the 5th term is 162.

Find the common ratio and the first term.

$$\frac{u_5}{u_2} = r^3 = \frac{162}{6} = 27 \quad (M1) \text{ A1}$$

$$r = 3 \quad \text{A1}$$

$$u_1 = \frac{u_2}{r} = \frac{6}{3} \quad (M1)$$

$$u_1 = 2 \quad \text{A1}$$

[4 marks]